

Evidence-Aware MapReduce for Forkable Compute

Separating Execution Fan-Out from Independent Evidence

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Abstract

Snapshot-backed sandboxes make branching cheap while leaving evidence dependence unchanged. Branches can reuse a model, prompt, repository, tests, observations, or execution ancestor, so counting outputs can amplify one repeated error into high-confidence consensus. We introduce an *evidence-aware reduction contract*: each worker reports an estimate, estimated information, evidence identifiers, fork lineage, and execution metadata. For independent workers estimating one common parameter, we use standard inverse-information pooling in its Gaussian/Wald form. The fixed-dimensional numeric summary can merge in any tree order; evidence IDs and lineage follow separate rules. The residual Δ measures disagreement, becomes Cochran’s Q in the scalar inverse-variance case, and appears in the product integral. A reference implementation validates serialized records, rejects repeated nonempty evidence identifiers, carries evidence and lineage through tree reduction, and uses Cholesky-based numerical linear algebra. Unit tests and seeded synthetic checks exercise the algebra, unequal information, and forged precision; one four-worker named-snapshot trace exercises the end-to-end path. Platform logs document the exercised execution paths. A central open systems challenge is to turn evidence identity and fork lineage into a dependence model for correlated and adaptively selected AI branches. Artifact: github.com/zozo123/boltzmann-mapreduce.

1 Introduction

Execution fan-out and evidence fan-out are different resources. A snapshot runtime can restore one prepared environment into many isolated workers, but shared models, prompts, tests, data, repositories, and ancestors can induce common-mode error. If K scalar estimates have equal marginal variance σ^2 and exchangeable pairwise correlation ρ , then

$$\text{Var}(\bar{\theta}) = \sigma^2 \left[\rho + \frac{1 - \rho}{K} \right], \quad (1)$$

whereas an independence calculation reports σ^2/K . For fixed $\rho > 0$, variance has the floor $\rho\sigma^2$ even as the number of branches grows. Execution isolation constrains runtime state; ρ instead depends on the evidence sources and branch history.

MapReduce gained leverage from a small interface that let the runtime manage partitioning, placement, retries, and communication [4]. Modern evaluation and agent

pipelines often retain the fan-out but reduce each branch to a scalar or vote. Such outputs erase four facts a reducer needs: uncertainty, information strength, evidence identity, and inherited dependencies. Equal voting ignores the first two; repeated evidence and shared ancestry can manufacture false precision from the latter two. Snapshot and serverless systems make this failure mode operationally important because branching prepared state is increasingly convenient [1, 14, 5].

We ask a systems question: *what should a worker return so that a runtime can reduce its result without erasing the evidence behind it?* Our answer is a structured worker record and a mergeable canonical state. The implementation covers the independent common-target case, rejects literal reuse of caller-supplied evidence identifiers, and carries lineage into the final result. Turning that lineage into a dependence estimate is the next research step.

Contributions.

- **Interface.** An evidence-carrying worker record and a fixed-dimensional Gaussian numeric state for flat, streaming, and tree reduction, with provenance aggregation and exact duplicate-ID rejection.
- **Artifact.** A reference implementation with record validation, closed-form product-integral evaluation, Cholesky solves, deterministic statistical checks, and a snapshot-backed integration trace.
- **Scope and agenda.** A precise separation between execution isolation and independent evidence, exposing correlation inherited through fork DAGs, confidence calibration, adaptive selection, and forged precision as reducer requirements.

2 Evidence-Aware Worker Results

Wire record. Worker k emits

$$r_k = (\hat{\theta}_k, J_k, n_k, \mathcal{E}_k, \mathcal{L}_k, m_k), \quad (2)$$

where $\hat{\theta}_k \in \mathbb{R}^p$ is an estimate, J_k is estimated per-observation information, n_k is a literal sample count, $\mathcal{E}_k$ contains evidence identifiers, $\mathcal{L}_k$ records fork lineage, and m_k stores execution metadata. The Python dataclass has a JSON-compatible serialized form. At ingestion, the implementation rejects non-finite estimates, non-positive integer counts, malformed provenance, inconsistent dimensions, and non-symmetric or non-positive-definite J_k . The current contract requires positive definiteness; the algebra also admits semidefinite local factors when their aggregate precision is full rank.

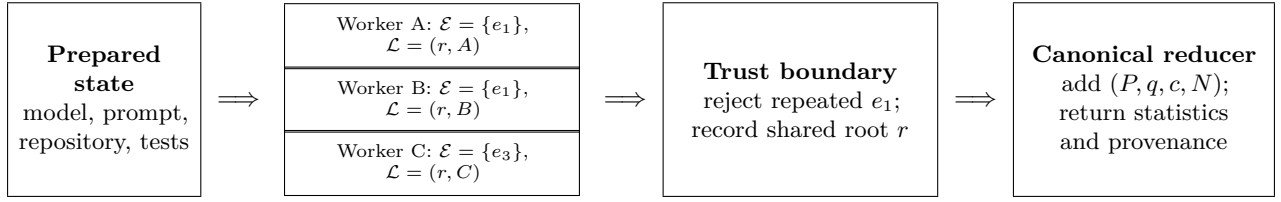


Figure 1: Separate workers can still reuse evidence. Workers A and B declare the same evidence e_1 , so the exact-overlap guard rejects their joint reduction. The shared root r remains in lineage when identifiers differ, preserving a signal that a future correlation model could use.

The invariant quantity is total precision

$$P_k = n_k J_k, \quad q_k = P_k \hat{\theta}_k, \quad c_k = \hat{\theta}_k^T P_k \hat{\theta}_k. \quad (3)$$

The decomposition $P_k = n_k J_k$ is useful when n_k is an actual observation count; workers with bootstrap, clustered, weighted, or verifier-derived uncertainty may instead require a direct precision field and a separate evidence-mass convention.

Trust semantics. Evidence identifiers and lineage are optional caller-supplied declarations whose reliability rests on the caller. The reducer compares nonempty $\mathcal{E}_k$ sets and rejects exact overlap by default. This check covers literal ID reuse; empty identifiers, relabeling the same data, and shared noise fall outside it. The record carries dimensions but no validated estimand, unit, model-version, or coordinate identifier. A production runtime must bind these fields to authenticated data and model manifests before treating equal dimensions as a common target.

Numeric merge and provenance. Independent summaries add

$$(P, q, c, N) = \bigoplus_k (P_k, q_k, c_k, n_k). \quad (4)$$

In exact arithmetic the numeric state is associative and commutative, enabling arbitrary tree reduction. The scalar c must survive intermediate merges: re-emitting only $(\hat{\theta}, J, N)$ would erase disagreement accumulated below the tree. Provenance follows its own semantics. On merge, evidence IDs are canonicalized as a set; lineage uses order-preserving deduplication, so the full serialized summary is order-sensitive. The final result returns both provenance fields. Numeric state costs $O(p^2)$, whereas an exact provenance ledger can grow as $O(|\mathcal{E}| + |\mathcal{L}|)$.

Relationship to sufficient-statistic reduction. Model-specific sufficient statistics remain preferable when available: a mean is recovered from $(\sum_i x_i, n)$ and ordinary least squares from $(X^T X, X^T y)$. The contract targets one-shot settings where workers expose estimates and calibrated uncertainty but raw data or model-specific sufficient statistics are unavailable or undesirable to transmit.

Execution backends. The same record works across processes, containers, and microVMs. The local backend uses Python’s platform-dependent process pool. The documented islo workflow prepares and saves a named environment before the normal backend restores one sandbox per shard. This separation lets an execution planner choose placement and tree shape without changing the statistical contract.

Retries and speculative execution. An evidence token identifies an evaluation independently of the process that executes it. A retry after failure and a speculative duplicate should therefore reuse the original evidence token; accepting both would turn fault tolerance into double counting. The current overlap guard raises an error when it sees a repeated token. A production keyed admission layer should retain one result for retries or speculative duplicates. A branch that draws a genuinely new sample receives a new token while retaining its parent lineage. Production tokens should bind the input partition, evaluator, model and prompt versions, and selection context through authenticated structured records.

3 Associative Gaussian Reduction

Statistical scope. Confidence distributions and inverse-information methods combine evidence from independent sources [12, 16, 17]; Rao-CD places estimating-function inference in a MapReduce setting [19]. We use their Gaussian/Wald common-effect special case. Assume disjoint evidence with independent estimation noise, a common parameter θ_0 , fixed p and K , $\min_k n_k \rightarrow \infty$, and calibrated positive-definite precision estimates. Write V_k for worker k ’s per-observation asymptotic covariance and suppose

$$n_k^{1/2}(\hat{\theta}_k - \theta_0) \xrightarrow{d} N_p(0, V_k), \quad J_k \xrightarrow{p} V_k^{-1}. \quad (5)$$

For efficient likelihood workers, local asymptotic normality is one route to this display; for estimating-equation workers, V_k is a sandwich covariance and V_k^{-1} is Godambe precision [7]. The leading Wald confidence kernel is

$$g_k(\theta) = \exp\left\{-\frac{1}{2}(\theta - \hat{\theta}_k)^T P_k(\theta - \hat{\theta}_k)\right\}. \quad (6)$$

It is exact when the local estimate is Gaussian with fixed known covariance. With plug-in covariance or non-Gaussian errors, it is a first-order asymptotic approximation.

Pooled center and heterogeneity. Let $P = \sum_k P_k$, $q = \sum_k q_k$, and $c = \sum_k c_k$. Completing the square gives

$$\hat{\theta} = P^{-1}q, \quad \Sigma_g = P^{-1}, \quad (7)$$

where Σ_g is exactly the covariance of the normalized product kernel and, under Eq. (5), a first-order estimate of the sampling covariance of $\hat{\theta}$. The residual quadratic is

$$\begin{aligned} \Delta &= \sum_k (\hat{\theta}_k - \hat{\theta})^T P_k (\hat{\theta}_k - \hat{\theta}) \\ &= c - q^T P^{-1} q \geq 0 \quad \text{in exact arithmetic.} \end{aligned} \quad (8)$$

For $p = 1$, Δ is Cochran’s Q ; the vector expression is the corresponding Gaussian weighted-residual diagnostic with direct multivariate meta-analytic analogues [3, 8]. Under independent $N_p(\theta_0, P_k^{-1})$ estimates with fixed known P_k , $\Delta \sim \chi_{p(K-1)}^2$; plug-in precision makes this calibration asymptotic. Small values remain compatible with dependence or duplicated evidence. Comparisons across different K must account for degrees of freedom.

For the unit-height kernels in Eq. (6), the partition integral is

$$Z_g = (2\pi)^{p/2} |P|^{-1/2} \exp(-\Delta/2). \quad (9)$$

Thus $\Delta/2$ is both the minimized summed quadratic energy and the heterogeneity contribution to $-\log Z_g$. The full integral also depends on $|P|$ and the integration measure. Its magnitude alone cannot be read as a conflict score or model evidence. Normalized local kernels require their own Gaussian constants.

Numerics and naming. The implementation obtains $\hat{\theta}$, Σ_g , and $\log |P|$ from a Cholesky factor, avoiding explicit matrix inversion. It evaluates Eq. (9) in closed form, clips a negative computed Δ only within a scale-aware roundoff tolerance, and rejects larger violations of the canonical invariant; the subtraction can otherwise suffer cancellation at large offsets. Writing

$$E_k(\theta) = \frac{1}{2}(\theta - \hat{\theta}_k)^T J_k(\theta - \hat{\theta}_k), \quad \beta_k = n_k \quad (10)$$

makes $g_k = \exp\{-\beta_k E_k\}$ an unnormalized Gibbs kernel. This is a naming convention: n_k can be absorbed into J_k , and P_k is the invariant. A global $\beta = N = \sum_k n_k$ with weighted-average energy is the same product. Bounded local sample sizes leave the Gaussian approximation unjustified, and accumulated local bias can dominate when K grows [18, 19].

Limits of reported precision. Holding reported precisions fixed, worker o has local influence

$$\frac{\partial \hat{\theta}}{\partial \hat{\theta}_o} = P^{-1} P_o. \quad (11)$$

Low relative precision in a direction limits leverage there, but distance leaves the derivative unchanged; fabricated

high precision creates another way to control the pooled estimate. The artifact’s stress-test clip caps determinant-based precision volume and applies a coordinate-wise median/MAD redescending location weight. A determinant cap leaves the largest directional eigenvalue unbounded, and the location rule depends on the coordinate system. This heuristic instruments one failure mode without providing a Byzantine robustness guarantee.

4 Evaluation of the Reduction Contract

The evaluation separates unit-tested identities, seeded behavioral checks, and execution traces. All statistical data are synthetic. Table 2 states the conclusion supported by each check. The point gaps summarize these runs; estimating statistical efficiency would require a broader repeated-sampling study.

Algebra and local estimators. Twelve homoscedastic mean shards of size 107–355 produce 4.9542 with a nominal plug-in 95% Wald interval [4.8980, 5.0104]; the full-sample mean is 4.9546. The reducer agrees with closed-form inverse-variance pooling to floating-point precision, and flat and tree reductions agree. This checks algebra; $(\sum x, n)$ would recover the sample mean exactly. In the linear-regression check, the pooled coefficient has norm gap 0.033 to centralized OLS. Each worker estimates its own residual variance, whereas exact aggregation of $X^T X$ and $X^T y$ uses a different centralized variance treatment.

Nonlinear and adversarial checks. Five IID logistic shards have sizes $\{60, 120, 400, 2000, 5000\}$. Across eight fixed seeds, information pooling is 0.0083 ± 0.0042 from the centralized MLE, while equal coefficient averaging is 0.177 ± 0.090 away. We use unweighted averaging as a simple baseline; a broader study would also compare sample-size weighting, Rao-CD, surrogate likelihood, and distributed Newton. In the forged-precision check, one scalar worker reports a distant location from a 2,000-point shard and inflates its reported precision by a further $50\times$; unprotected pooling moves to 17.0004, and the stress clip returns 4.9566. This check exposes the vulnerability and illustrates one narrow response.

Execution paths. One integration trace starts from a named 141 MB *islo* snapshot and restores four sandboxes. Their seed-pinned workers return pooled mean 4.9422 versus full-sample 4.9450; four concurrent restore–run–capture round trips complete in 6.70 s. This validates the snapshot-to-worker-to-reducer path with client-observed timing.

Table 1 records the operation exercised on each platform. Higher requested concurrency reached validation or capacity limits for Daytona at 32 and *islo* at 16 and 32.

5 Assumptions and Open Problems

Dependence. Literal ID deduplication catches exact declared reuse. A lineage union records common ancestors

Table 1: Platform-specific batched traces. Daytona and Tensorlake percentiles cover sandbox creation; their total wall time covers the complete create-run-delete batch. The islo percentiles cover restore-run-capture round trips.

Platform	State source	Measured operation	Client conc.	Batch	Success	p_{50}	p_{95}	Total wall
Daytona	default environment	create	8	1024	1024/1024	0.20 s	1.12 s	190.3 s
Tensorlake	default environment	create	8	256	256/256	3.44 s	9.00 s	387.3 s
islo	named 141 MB snapshot	restore + run + capture	12	256	255/256	6.87 s	9.04 s	240.3 s

Table 2: Artifact checks and their supported claims. The logistic values are mean $\pm$ SD over eight seeds.

Check	Observation	Supported conclusion
Canonical algebra	Flat/tree and closed-form results agree	Code matches the stated Gaussian formulas
Evidence overlap	Repeated nonempty ID raises an error	Exact labeled reuse is blocked
Unequal-size logistic	pool gap 0.0083 ± 0.0042 ; equal average 0.177 ± 0.090	Information pooling improves on unweighted averaging in this check
Forged precision	attacked 17.0004; clipped 4.9566	Vulnerability and behavior of one heuristic
Snapshot integration	four workers complete in 6.70 s	Named-snapshot path executes end to end

without encoding a fork DAG or mapping ancestry to covariance. A correlation-aware reducer should combine factors conditionally on shared observations, evaluators, model state, and selection history. Equation (1) makes this a central statistical problem.

From lineage to a fork-DAG reducer. A useful next record would name parent branch IDs, immutable evidence manifests, model/prompt/tool hashes, evaluator and test versions, and branch-selection events. The runtime could then map known overlap to shared latent factors, covariance blocks, or conservative evidence groups before applying a generalized reducer. Crucially, the interface must permit abstention: when the runtime cannot justify a dependence model, it should report unresolved correlation and withhold a narrower interval. The current lineage union records the history while leaving this modeling step open.

Different local targets. The common- θ_0 assumption defines a common-effect model. Non-IID observations can still support pooling when every worker identifies the same parameter and reports valid uncertainty. If workers target different estimands, P^{-1} omits between-worker variation and can be severely overconfident. The system must first name the target and adopt an explicit heterogeneity model, such as site effects, meta-regression, or a random-effects hierarchy.

Confidence construction and adaptive selection. A model’s verbal confidence requires empirical calibration before it can serve as a standard error; calibration depends on the model, elicitation, task, and deployment

distribution. Agent precision should be derived from external evidence such as held-out tests, repeated trials, independent verifiers, or an empirically calibrated outcome model. If the same evidence selects surviving branches and constructs their factors, winner’s-curse bias can result; the ledger must record that reuse.

Untrusted workers and systems evidence. Robust aggregation requires a threat model over locations, covariance reports, collusion, and dimension. Controlled systems comparisons likewise require identical images and workloads, concurrent fan-out, tail latency, memory sharing, failures, cost, and statistical utility per unit time. The present artifact supplies interfaces and traces on which those studies can be built.

6 Related Work

The Gaussian reducer follows established confidence-distribution and common-effect meta-analysis [12, 16, 3]. Rao-CD and distributed GLM inference provide closely related estimating-equation constructions [19, 13]. One-shot averaging and surrogate likelihood explore neighboring communication-statistics tradeoffs [18, 9]. We add a runtime contract that makes the independence conditions visible before reduction.

Self-consistency marginalizes answers from diverse reasoning paths, while multiagent debate lets models exchange critiques before converging [15, 6]. These methods exploit redundant generations for answer selection while leaving evidence dependence unstated. Our contract records provenance, and its Gaussian reducer combines uncertainty when the independence assumptions hold.

MapReduce established the programmable dataflow contract [4]; Firecracker, Catalyzer, REAP, and Faasm study lightweight isolation, startup, and reusable state [1, 5, 14, 11]. These systems make branching practical; our work connects that capability to explicit evidence accounting at reduction time. Robust distributed learning methods such as Krum and FLAME start from explicit adversary models [2, 10]. The included stress clip is a diagnostic heuristic without a Byzantine-robustness guarantee.

7 Conclusion

Cheap forks reduce execution cost while leaving the amount of independent evidence unchanged. The contract developed here makes an estimate, uncertainty, evidence identity, and lineage explicit, then reduces independent Gaussian summaries through an associative numeric state while retaining provenance. The reference artifact vali-

dates this path and reports residual heterogeneity explicitly. The next systems step is a reducer over the fork DAG: one that can explain why two branches count as separate evidence, model the dependencies they inherited, and remain calibrated after selection. Faster fan-out is useful; evidence-aware fan-in keeps the assumptions and unresolved dependencies visible.

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